

Sammy Abdella

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New York City, NY

Computer Scientist

I am a results-driven software engineer with a passion for backend systems, data optimization, and scalable solutions. My background spans financial institutions, postal logistics, and academic research, with a consistent record of improving system efficiency, ensuring compliance, and enabling better decision-making through data.

Areas of Expertise

Backend Development • API Optimization • Microservices Architecture • Data Analysis • PostgreSQL Optimization • OAuth2/JWT • Docker & Kubernetes • Scientific Computing • Medical Software Tools • Research & Technical Writing • Multilingual Communication (English, French, Arabic)

Education

Bachelor of Science in Computer Science | Columbia University – New York City, NY

May 2027

Professional Work History

Web Developer Intern | Columbia College Information Technology

May 2025 – Present

- Migrated 15 Columbia departmental websites from Drupal 7 to Drupal 10, writing migration scripts, refactoring custom modules, and handling regression testing, improving average page load times by ~35% and eliminating 100% of deprecated modules in production
- Integrated a Django, PostgreSQL, and RESTful API backend with a new Columbia admissions website, scaling reliably to 50,000+ applicants per cycle, cutting form submission errors by ~40%, and reducing manual data correction work by ~50%

Research | Columbia University Artificial Intelligence Laboratory

January 2025 – September 2025

- Lead engineer for gAyl, an AI tool for LGBTQ+ suicide prevention. Designed six model-based safety benchmarks to generate treatment plan recommendations and achieved <15% false-negative rate on unsafe-response detection across 1,000+ high-risk prompts
- Trained LLM-based classifiers in PyTorch and Hugging Face Transformers to automate safety evaluation, reducing manual benchmarking time by ~95%
- Publication: <https://doi.org/10.7916/z6vc-d867>

Software Developer Intern at Zitouna Bank | Tunis, Tunisia – Remote

January 2024 – May 2024

- Engineered and optimized RESTful APIs using Python and Django, cutting response times by 30% and boosting transaction throughput by 25%
- Partnered with database teams to manage and tune PostgreSQL performance for real-time, high-volume financial data
- Ensured backend infrastructure aligned with security standards and financial regulations, resulting in a 100% compliance rate during audits

Freelance Software Developer | Fiverr – Remote

November 2021 – December 2021

- Assisted 50+ clients in customizing web applications, developing mobile apps, integrating APIs, optimizing website performance, and implementing cybersecurity measures, \$10,000+ total earnings

Technical Skills

Python, C++, Java, JavaScript, SQL, HTML/CSS, Django, RESTful APIs, OAuth2, JWT, Docker, Kubernetes, PostgreSQL, API Integration, Web Optimization, Kubernetes, Large Language Models (LLMs), Jupyter Notebook, Matplotlib, Scientific Computing, PostgreSQL Performance Tuning, SQL Query Optimization, Database Indexing, API Performance Optimization, Microservices Architecture, CI/CD Pipelines, Unit Testing, Regression Testing, Content Management Systems (CMS), Drupal 7–10 Migrations, Website Performance Optimization, Security & Compliance (Backend Systems)